Varun Madabushi

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Overview

3rd-year PhD student seeking an internship in the robotics field. Researching robot locomotion, learning, and control, with a wide breadth of skills spanning hardware and software domains. Interested in positions involving physics simulation, robot design, and control of safety-critical systems.

Education

Georgia Institute of Technology — Atlanta, GA

B.S. in Electrical Engineering, Minor in Robotics (GPA: 3.93/4.00) — *Aug 2017 – Dec 2020*

M.S. in Electrical and Computer Engineering (GPA: 3.92/4.00) — *Jan 2021 – May 2022*

PhD. in Robotics (Advised by Dr. Maegan Tucker) — *Aug 2024 – May 2029 (Expected)*

Research Experience

Robot Learning for Locomotion

Advised by Dr. Maegan Tucker — *Aug 2024 – Present*

- Researching learning-based approaches to co-design the morphology and controller across a class of robots and locomotion tasks. Presented preliminary results at IROS 2026
- Developed NaviGait, a bipedal robot control system utilizing offline trajectory optimization and reinforcement learning to generate stepping behaviors that are natural-looking, easily tunable, and train faster than comparable approaches (presented at ICRA 2026).
- Assisted development of humanoid motion regtargeting algorithm which integrates motion capture and ground reaction force data alongside trajectory optimization to produce dynamically feasible and human-like reference trajectories.

Reachability Analysis of Hybrid Systems

Advised by Dr. Maegan Tucker, Dr. Samuel Coogan — *Aug 2024 – Present*

- Studying safety verification and robust control of hybrid systems using fast computation of forward-reachable sets, with applications to safe locomotion of legged robots.
- Developed a criteria for verifying forward invariance of a hybrid system's forward reachable set, and applied this to design a maximally robust controller for a bipedal walker model (Accepted to CDC 2026).
- Created algorithm for probabilistic invariant set computation through sampling-based optimization and PAC learning for correctness guarantees.

Planning and Control for Unmanned Aerial Vehicles

Advised by Dr. Patricio Vela — *Jan 2022 – Aug 2022*

- Developed kinodynamic trajectory planner in MATLAB using nonlinear optimization to generate composite bezier spline paths satisfying dynamic constraints for fixed-wing Unmanned Aerial Vehicles (UAVs).
- Combined model predictive and reinforcement learning-based elements in heirarchial aircraft controller to improve generalization to various aircraft types

Employment

Johns Hopkins University Applied Physics Laboratory — Laurel, MD

Robotics Engineer — *Jun 2022 – Jul 2024*

- Led statistical verification and operational analysis, and assisted in development of Nonlinear Model Predictive Control (NMPC) algorithm for multi-agent fixed-wing UAV systems (presented at ICRA 2025)
- Provided hardware and software support for outdoor flight tests of NMPC-based UAV landing controller
- Implemented trajectory optimization to realize manipulation behaviors such grasping and valve turning on unmanned underwater mobile manipulator (awarded top prize in internal R&D across all of the company)
- Co-led research on robot co-design, which leveraged Bayesian optimization to automatically choose task-optimal hardware and control parameters for hexapod robot system
- Designed 3-axis sensor positioning mechanism for littoral zone exploration robot
- Developed power and control electronics for thermal cautery heater used in experimental emergency medical device

- Mentored high school interns on mechanical and software design of robotic manipulator

Lawrence Livermore National Laboratory Livermore, CA

Defense Technologies Engineering Intern *May 2020 – Aug 2020, Sep 2021 – Apr 2022*

- Designed simulator and hardware-in-the-loop testing for Unmanned Surface Vehicle (USV) control system
- Developed actuator and hydrodynamic drag models fit from experimental data, used to inform new USV design
- Led development of data collection system integrating cameras, hydrophones, and FPGA on custom PCB

Publications

[1] **V. Madabushi**, A. Harapanahalli, S. Coogan, and M. Tucker, "Differentiable invariant sets for hybrid limit cycles with application to legged robots," in *IEEE CDC*, 2026.

[2] N. Janwani, **V. Madabushi**, and M. Tucker, "Navigait: Navigating dynamically feasible gait libraries using deep reinforcement learning," in *IEEE ICRA*, 2026.

[3] X. Zhang, S. Haener, **V. Madabushi**, and M. Tucker, "Kinodynamic motion retargeting for humanoid locomotion via multi-contact whole-body trajectory optimization," *Under Review*, 2026.

[4] **V. Madabushi**, Y. Kopel, A. Polevoy, and J. Moore, "Dense fixed-wing swarming using receding-horizon nmpc," in *IEEE ICRA*, 2025.

[5] R. Kim, **V. Madabushi**, E. Dong, and A. Mazumdar, "Increasing mobile robot efficiency and versatility through manipulation-driven adaptation1," *Journal of Mechanisms and Robotics*, vol. 13, no. 5, May 2021.

[6] **V. Madabushi**, E. Dietrich, H. Krasowski, and M. Tucker, "Finite-step invariant sets for hybrid systems with probabilistic guarantees," *Under Review*, 2026.

[7] **V. Madabushi**, N. Janwani, and M. Tucker, "Mocha: Multi-objective co-design using hypernetwork architectures," in *Under Review*, 2026.

[8] S. Bajaj, V. S. Madabushi, M. Tucker, and V. Gupta, "Comparison of non-deterministic nonlinear systems," in *Under Review*, 2026.

Skills

Concepts: Nonlinear Systems, Optimal Control, System Identification, Reinforcement Learning, Parallel Programming

Programming: C++, Python, JAX, PyTorch, CUDA, MPI, MATLAB

Software: MuJoCo, ROS, Docker, STM32, Arduino, Git, Solidworks, OnShape, Altium, KiCAD

Tools: Soldering (SMD and THT), Mill, Lathe, 3D Printer, Oscilloscope, Logic Analyzer

Teaching

ECE 4550 (Control System Design): Supervised lab assignments and gave supplemental lectures for undergraduate course covering linear controller and observer design, motors and power converters, and embedded systems.

ECE 6552 (Nonlinear Systems): Hosted office hours and review sessions for graduate course covering fundamentals of nonlinear control including Lyapunov theory, feedback linearization, and control barrier functions.

VIP (Experimental Bipedal Robotics): Mentored multidisciplinary team of 12 undergraduates developing a custom bipedal robot.